

PRANAV VISWANATHAN

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EDUCATION

Northeastern University

Master of Science, Computer Science

- Coursework: Machine Learning, MLOps, Algorithms, DBMS, Natural Language Processing, Building Scalable Distributed Systems

Boston, MA

Sept 2024 – May 2026

Vellore Institute of Technology

Bachelor's in Technology, Computer Science and Engineering

Chennai, India

2024

WORK EXPERIENCE

NergyLive

SDE Intern

May 2023 – Jul 2023

- Built real-time HVAC anomaly detection using isolation forests and statistical process control, processing 10K+ sensor readings/minute with sub-100ms latency in Python
- Deployed anomaly detection system into production with cross-functional teams, reducing downtime incidents and cutting maintenance costs by 15%; built Plotly dashboards enabling engineers to identify root causes 3x faster

PROJECTS

OmniRAG | Python, FastAPI, LlamaIndex, LanceDB, Next.js, React, Docker, Ollama

Feb 2026 – Mar 2026

- Built a local-first Retrieval-Augmented Generation system with a FastAPI backend and Next.js frontend, enabling fully private on-device LLM inference via Ollama with session-isolated vector persistence in LanceDB
- Implemented BGE cross-encoder reranking with all-MiniLM-L6-v2 SentenceTransformer embeddings, improving retrieval precision through cosine similarity scoring across semantically chunked document collections
- Designed smart fallback pipeline that routes low-confidence queries to an async web scraper (BeautifulSoup4, HTTPX, Wikipedia API) and built an interactive Semantic Knowledge Graph using react-force-graph-2d for cross-document relationship visualization with live citation tracking

Rebalance-AI: BlueBikes Demand Prediction | Python, Apache Airflow, MLflow, Docker, GitHub Actions

Sept 2025 – Dec 2025

- Engineered end-to-end ML pipeline processing 3.16M ride records with Apache Airflow orchestration, implementing automated feature engineering for weather, temporal, and geospatial data
- Developed production-grade ensemble models (XGBoost, LightGBM, Random Forest) achieving $R^2 > 0.96$, with MLflow integration for experiment tracking, model versioning, and deployment automation
- Built CI/CD pipeline using GitHub Actions for automated testing and Docker containerization, with comprehensive monitoring tracking model performance metrics across 400+ bike stations

Lucera: Advanced Video Processing Pipeline | Python, FFmpeg, Docker, PyTorch, Real-ESRGAN, Whisper

Dec 2025 – Jan 2026

- Engineered modular Python video processing pipeline integrating Real-ESRGAN 4x upscaling, RIFE frame interpolation (24fps to 60fps), and Whisper AI captioning with VMAF/PSNR quality metrics, maintaining industry-standard quality thresholds above 75 across 6 autonomous processing stages
- Architected loosely-coupled OOP classes orchestrating FFmpeg subprocess management, GPU batch inference, and adaptive HLS streaming with 4 bitrate variants, achieving sub-200ms frame analysis latency validated through 40+ pytest
- Containerized end-to-end pipeline using Docker with automated shell scripts for one-command deployment, enabling persistent artifact storage across denoising, interpolation, upscaling, and packaging stages

HawkEye | Python, OpenCV, PyTorch, Docker, Shell Scripting

Sept 2025 – Nov 2025

- Built real-time multi-sensor fusion pipeline processing synchronized RGB, thermal, and LiDAR streams at 14-16 FPS with sub-100ms end-to-end latency, implementing modular sensor adapters for independent stream ingestion and frame alignment
- Implemented Kalman Filter-based multi-object tracking with IoU-based Hungarian assignment, maintaining consistent track IDs across 400+ frames with automated state prediction, gating thresholds, and track lifecycle management
- Developed configurable sensor degradation simulation framework modeling fog attenuation, Gaussian noise injection, and signal jamming across all three modalities, enabling systematic robustness evaluation under 10+ edge-case scenarios

TECHNICAL SKILLS

Languages: Python, Java, Go, JavaScript, TypeScript, SQL, C/C++, R, Shell Scripting

AI & Machine Learning: RAG, Vector Embeddings, Cross-Encoder Reranking, Supervised/Unsupervised Learning, Reinforcement Learning, Ensemble Methods, Feature Engineering, Anomaly Detection

Frameworks & Libraries: FastAPI, Next.js, React, PyTorch, Scikit-learn, XGBoost, LightGBM, LlamaIndex, LanceDB, OpenCV, Hugging Face Transformers, SentenceTransformers, Pandas, NumPy

MLOps & Infrastructure: Docker, Apache Airflow, MLflow, GitHub Actions, Prometheus, Redis, Terraform, Git, Linux, pytest, Locust

Databases & Cloud: PostgreSQL, LanceDB, AWS (S3, EC2, RDS, Lambda, Fargate, SNS/SQS), Google Cloud Platform